Information Retrieval 5

Transformer and BERT for Ranking

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This lecture is based on the following textbook:

Jimmy Lin, Rodrigo Nogueira, Andrew Yates. *Pretrained Transformers for Text Ranking: BERT and Beyond*. Synthesis Lectures on Human Language Technologies, Springer, 2021.

Review of Previous Lecture: Traditional Retrieval Models

Boolean Model: Exact matching based on set theory. No ranking capability

Vector Space Model: TF-IDF weighting + cosine similarity. Weak theoretical foundation

Probabilistic Model: BIM based on PRP leading to BM25. Still the standard baseline today

Language Model: Probability of a document generating a query. Smoothing implicitly achieves IDF-like effects

In this lecture, we will learn the **Transformer / BERT** architecture and training techniques for ranking tasks that go beyond the limitations of traditional models

Gain a deep understanding of Transformer and BERT as the foundation of neural retrieval models, and learn training techniques for applying them to ranking tasks

First Half: From Neural Networks to Transformer and BERT

- Fundamentals of neural networks
- Word embeddings (distributed representations)
- Attention and Self-Attention
- Transformer architecture
- BERT pre-training and input representation

Second Half: Training for Ranking

- Basics of fine-tuning
- Loss functions for ranking
- Negative sampling strategies
- Additional pre-training and multi-step training
- MS MARCO training pipeline

Fundamentals of Neural Networks

Deep Learning for NLP

An **artificial neuron (perceptron)** generates an output by applying an activation function to the weighted sum of inputs

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

x_i: input, w_i: weight, b: bias, f: activation function

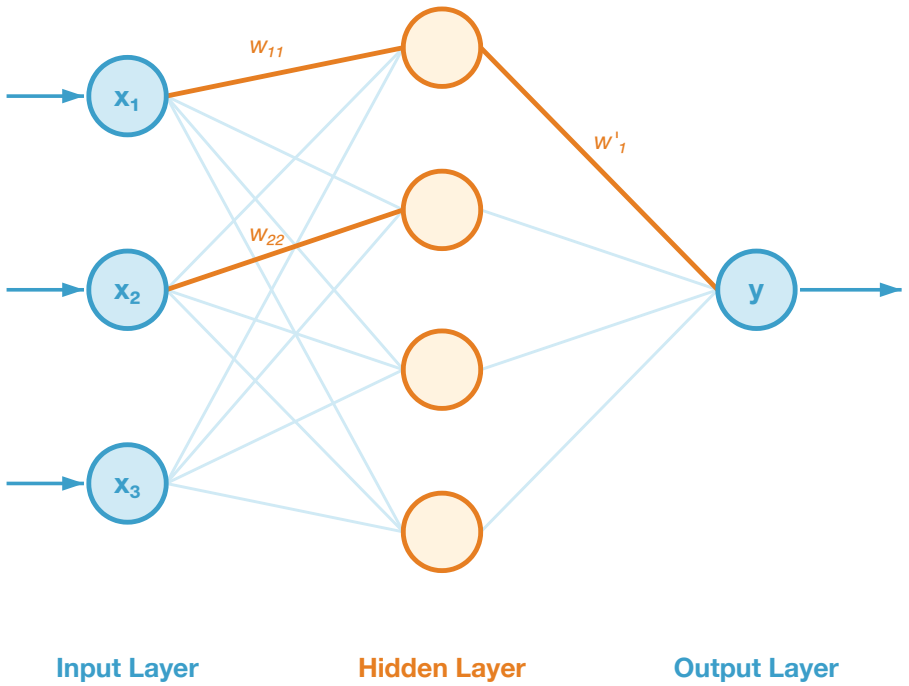
Weights w_i are **learned** from data (backpropagation)

A **multi-layer perceptron (MLP)** stacks multiple layers:

Input layer → **Hidden layer(s)** (one or more) → **Output layer**

Deeper layers can learn more complex patterns → **deep learning**

Multi-Layer Perceptron (MLP)



Each connection between nodes has a weight w that is learned from training data. The number of hidden nodes and layers are hyperparameters

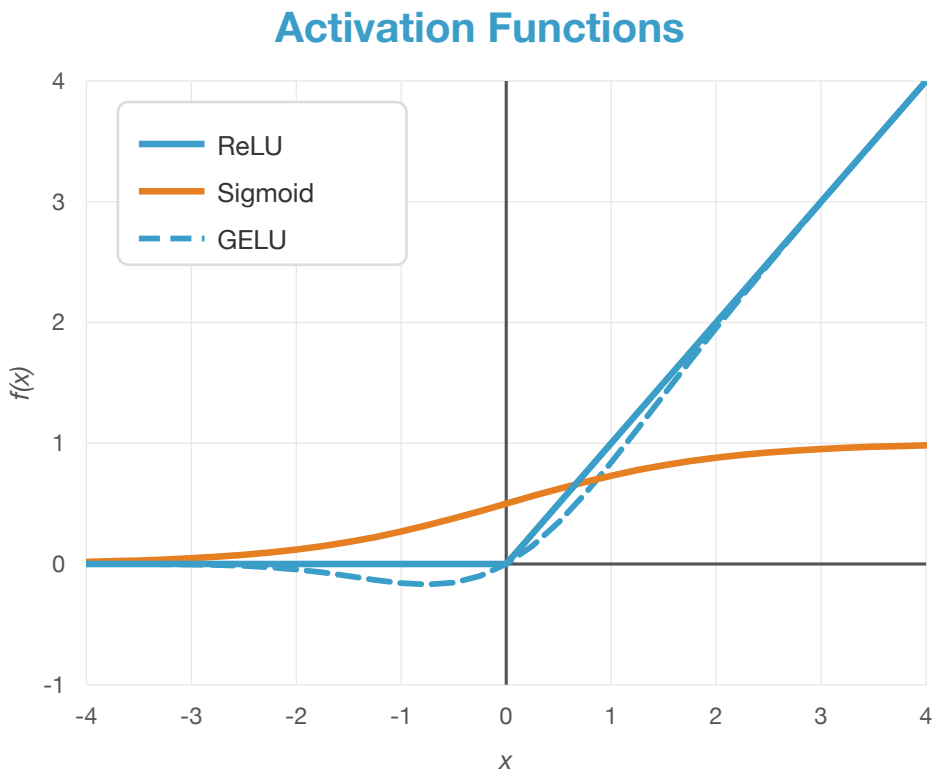
Without activation functions, even multi-layer networks are merely **compositions of linear transformations = a linear transformation**

$$f(Wx+b) \xrightarrow{f=\text{identity}} Wx+b$$

Non-linear activation functions enable approximation of increasingly complex functions as layers are stacked

Key activation functions:

- **ReLU**: $f(x) = \max(0, x)$ — Fast computation, less gradient vanishing. Standard in CNNs and MLPs
- **Sigmoid**: $f(x) = \frac{1}{1+e^{-x}}$ — Squashes output to $[0, 1]$. Used in output layers for binary classification
- **GELU**: $f(x) = x \cdot \Phi(x)$ — Adopted in BERT and GPT. A smooth approximation of ReLU
  - $\Phi(x)$: CDF of the standard normal distribution (x large → 1, small → 0)



The Feed-Forward layers in Transformer / BERT use **GELU**. It avoids the "discontinuity at 0" problem of ReLU while having a probabilistic interpretation. The GELU graph is shown overlaid with ReLU (orange dashed line) for comparison

Word Distributed Representations — Embedding

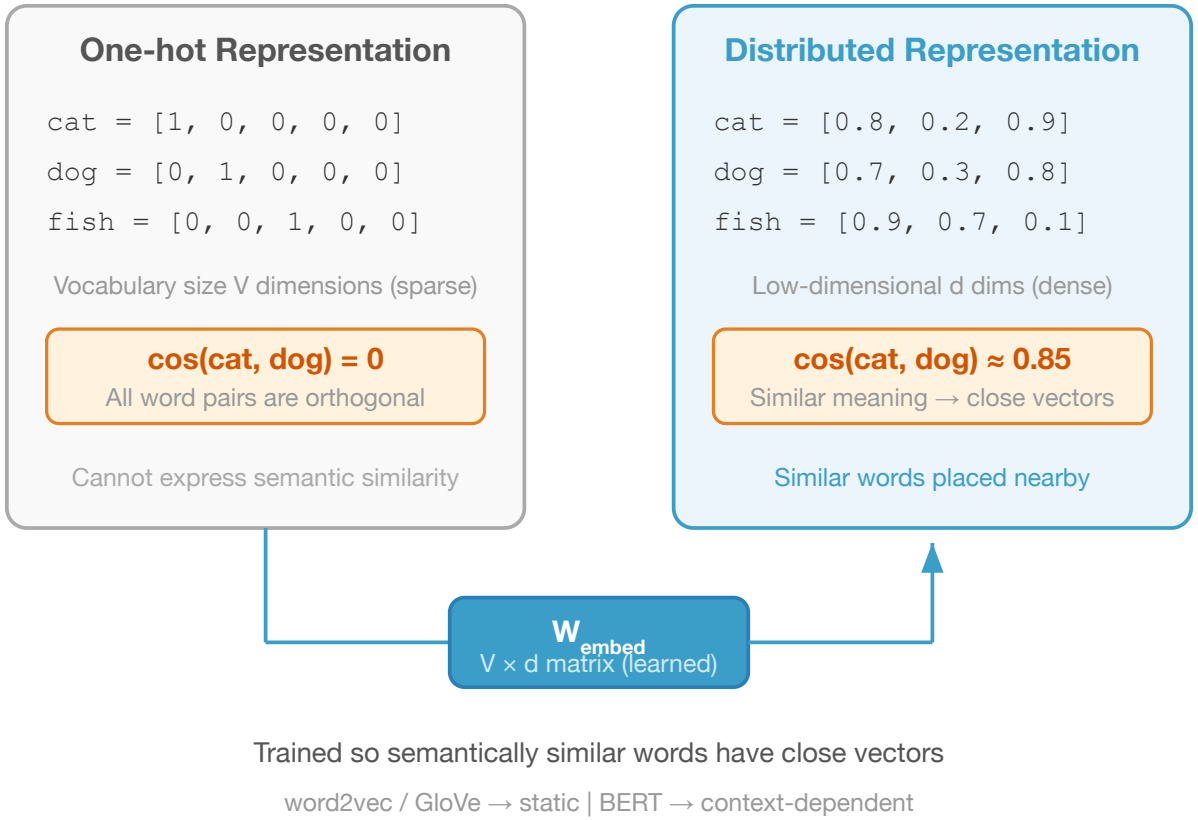
In NLP, words are represented as **fixed-dimensional real-valued vectors**. This is called an **embedding**

Problems with one-hot representation:

- 100K vocabulary → 100K-dimensional sparse vector
- **Cannot represent relationships between words**: cos("Kyoto", "Nara") = 0

Advantages of dense embeddings:

- Low-dimensional (100–768 dimensions) dense vectors
- Learned so that **semantically similar words have similar vectors**
- cos("Kyoto", "Nara") ≈ 0.85 (high similarity)



Trained so semantically similar words have close vectors

word2vec / GloVe → static | BERT → context-dependent

From word2vec to Contextualized Embeddings

word2vec [Mikolov et al., 2013]:

- Learns embeddings by predicting a target word from surrounding words (Skip-gram) or vice versa (CBOW)
- First large-scale realization of "semantic similarity ≈ vector proximity"
- **Limitation**: One vector per word → **Cannot distinguish polysemous words**

Contextualized Embeddings:

- Generate **different vectors for the same word depending on context**
- ELMo [Peters et al., 2018] → **BERT** [Devlin et al., 2019]
- Important for IR: Can distinguish "bank (financial)" from "bank (river)" by context

Static vs Contextualized Embeddings

word2vec (Static)

"bank in **account**" → $\mathbf{v}_{\text{account}}$
"river **account**" → $\mathbf{v}_{\text{account}}$ (same!)

BERT (Contextualized)

"bank account" → $\mathbf{h}_{\text{account}}^{(1)}$ financial context
"cooking account" → $\mathbf{h}_{\text{account}}^{(2)}$ different context
→ The two vectors are **different**

Each layer of BERT progressively integrates contextual information, producing sophisticated semantic representations in the final layer

Mikolov et al., "Efficient Estimation of Word Representations in Vector Space", ICLR, 2013; Peters et al., "Deep contextualized word representations", NAACL, 2018

The Idea Behind Attention — Learning "What to Focus On"

Attention originated in machine translation [Bahdanau et al., 2015]

Problem: Compressing a long sentence into a fixed-length vector with an RNN causes information from the beginning to be lost

Solution: At each step, the decoder **refers to all positions in the encoder** and **"attends to" positions with high relevance**

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})}$$

α_{ij}: attention weight that decoder position i places on encoder position j

e_{ij}: relevance score between positions i and j

Key insight: The model can **automatically** learn "which parts of the sequence are important" from data

Example of Attention in Translation

Input (English): "The cat sat on the mat"

Output (Japanese): "猫がマットの上に座った"

When generating "**猫**" (cat):

The **cat** sat on the mat

→ A **high attention weight** is assigned to "cat"

Self-Attention in the Transformer is an extension of this Attention idea, applied within the same sequence

Bahdanau et al., "Neural Machine Translation by Jointly Learning to Align and Translate", ICLR, 2015

Transformer

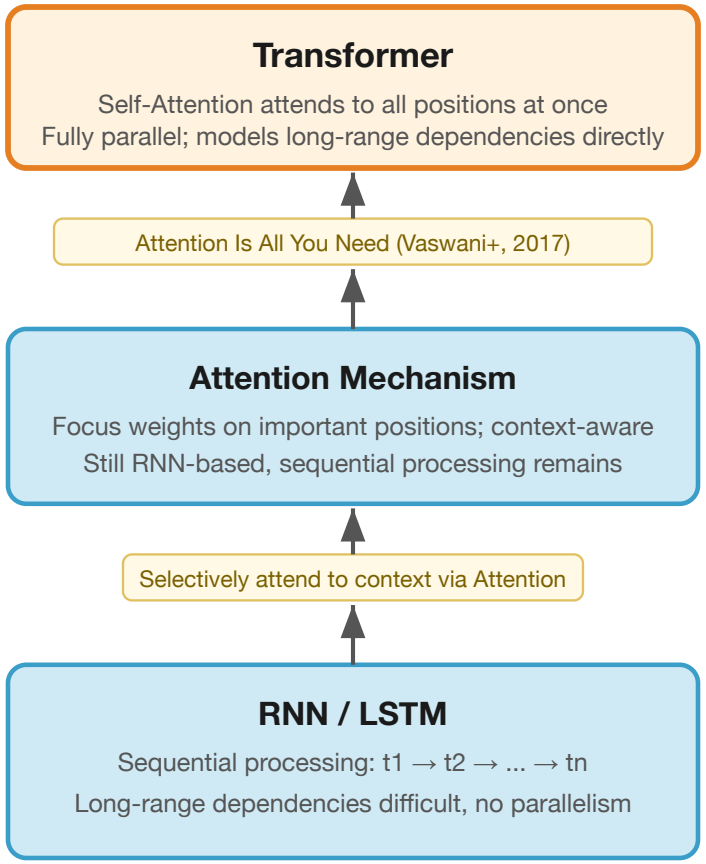
Why Is the Transformer Needed?

Limitations of previous models:

- **DSSM**: Independent encoding → no term interactions
- **DRMM**: Similarity matrix but no context
- **RNN/LSTM**: Sequential, gradient vanishing, hard to parallelize

Transformer [Vaswani et al., 2017]:

- **Self-Attention** captures full-sequence context simultaneously
- Parallel computation, directly models long-range dependencies



Computational complexity comparison:
RNN: O(n) steps for sequence length n (sequential)
Transformer: O(1) steps (parallel), but each step $O(n^2)$

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

Self-Attention is a mechanism where **each token in the input sequence computes its relevance to all other tokens**, generating context-dependent representations

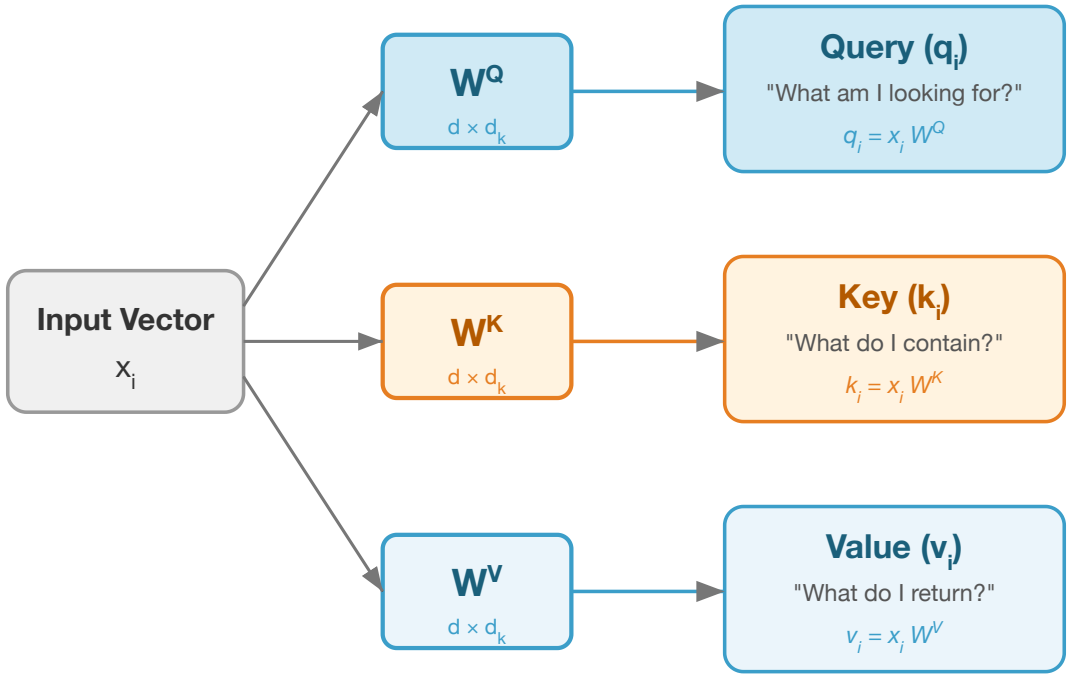
From each token's input vector x_i, three vectors are generated:

Q (Query): "What am I looking for?" — $q_i = x_i W^Q$

K (Key): "What do I have?" — $k_i = x_i W^K$

V (Value): "What do I return?" — $v_i = x_i W^V$

Intuition: Each token "asks a question (Q)" to other tokens, and aggregates "values (V)" according to the match with "keys (K)"



Each token independently produces Q, K, V (weight matrices shared across all tokens)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

Input: "Kyoto no koyo" (3 tokens, simplified example with $d_k = 2$)

Step 1: Generate Q, K, V from each token

	Q	K	V
Kyoto	[1, 0]	[0, 1]	[1, 1]
no	[0, 1]	[1, 0]	[0, 1]
koyo	[1, 1]	[1, 1]	[1, 0]

Step 2: Dot product of Q and K → score matrix

$$QK^T = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

Step 3: Scale by $\sqrt{d_k} = \sqrt{2} \approx 1.41$ and apply softmax

$$\frac{QK^T}{\sqrt{d_k}} \approx \begin{pmatrix} 0 & 0.71 & 0.71 \\ 0.71 & 0 & 0.71 \\ 0.71 & 0.71 & 1.41 \end{pmatrix}$$

Softmax of the "koyo" row: $[0.22, 0.22, 0.56]$

→ "koyo" **attends most to itself** while also referencing "Kyoto" and "no"

Step 4: Weighted sum of V using attention weights

$$\begin{aligned} \text{out}_{\text{koyo}} &= 0.22 \cdot [1, 1] + 0.22 \cdot [0, 1] + 0.56 \cdot [1, 0] \\ &= [0.78, 0.44] \end{aligned}$$

In actual BERT, d_k = 64, sequence length = 512, but the principle is the same

Why divide by $\sqrt{d_k}$?

The dot product of Q and K grows in scale as the dimensionality d_k increases

- Assuming each element is an independent random variable with mean 0 and variance 1:
  - The variance of $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ is proportional to d_k
  - For $d_k = 64$, the standard deviation of the dot product ≈ 8

Problem: When dot product values are large, softmax produces an **extreme distribution** (concentrated on one, others nearly 0)

$$\text{softmax}([10, 1, 1]) \approx [0.9999, 0.0001, 0.0001]$$

→ Gradients become nearly 0, preventing learning

Solution: Dividing by $\sqrt{d_k}$ normalizes the variance of the dot product to 1

Effect of Scaling

Without scaling (d_k = 64)

Dot products: [52.3, 3.1, -8.7]
softmax: [**1.000**, 0.000, 0.000]
→ Attends to only one token (hard attention)
→ Gradient vanishing makes learning difficult

With scaling (÷ $\sqrt{64}$ = 8)

After scaling: [6.5, 0.4, -1.1]
softmax: [**0.997**, 0.002, 0.001]
→ Smoother distribution
→ Gradients flow, learning stabilizes

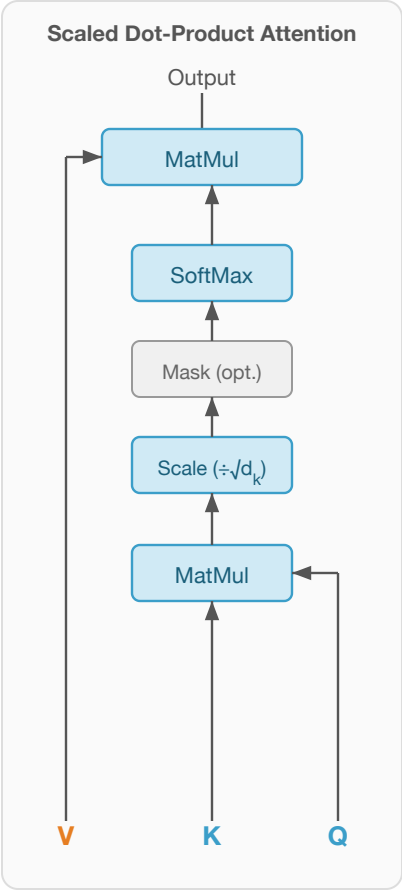
Multi-Head Attention

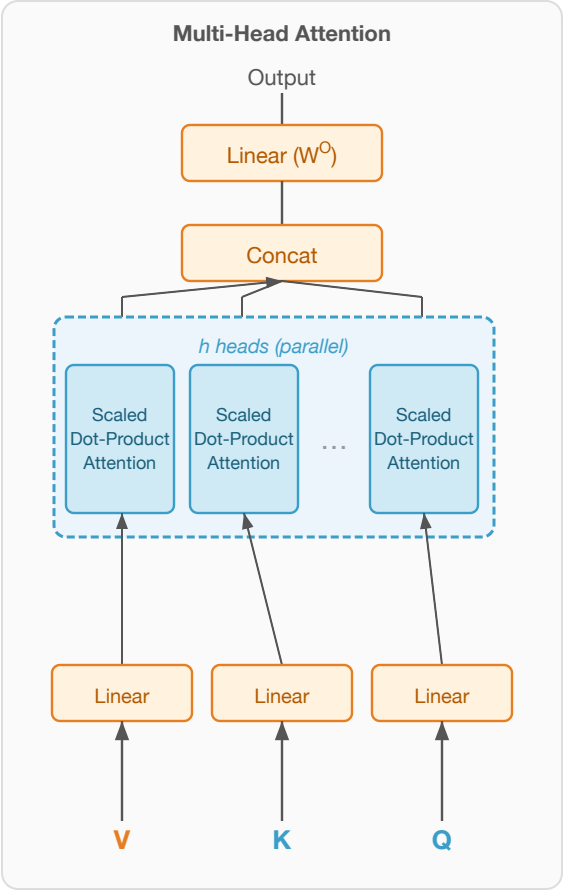
A single Attention can only capture one type of relationship → **Run multiple Attentions in parallel**

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W^O$$

$$\mathrm{head}_i = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

- h: Number of heads (BERT-Base: $h = 12$)
- Each head learns a **different type of relationship**
  - One head: Syntactic relationships (subject-predicate)
  - One head: Semantic relationships (synonyms)
  - One head: Positional proximity





Each head operates on $d_k = d_{model} / h = 768 / 12 = 64$ dimensions, so the computational cost is roughly the same as a single full attention head

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

Positional Encoding — Injecting Position Information

Self-Attention **does not distinguish input order** (set function) → need **Positional Encoding**

Original Transformer — Sinusoidal:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right), \quad PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

pos: position, i: dimension index. Different wavelengths per dimension

BERT — Learnable Position Embedding: learnable parameter per position (max 512)

Intuition for Positional Encoding

Input embeddings (without position info):

cat sat down

+ Positional encoding:

pos=0 pos=1 pos=2

= Position-aware embeddings:

cat@0 sat@1 down@2

Sinusoidal vs learnable achieve similar performance, but BERT cannot exceed 512 tokens

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017; Devlin et al., 2019

Residual Connection and Layer Normalization

Two techniques to prevent **gradient vanishing** in deep networks:

Residual Connection: $\text{output} = x + \text{SubLayer}(x)$

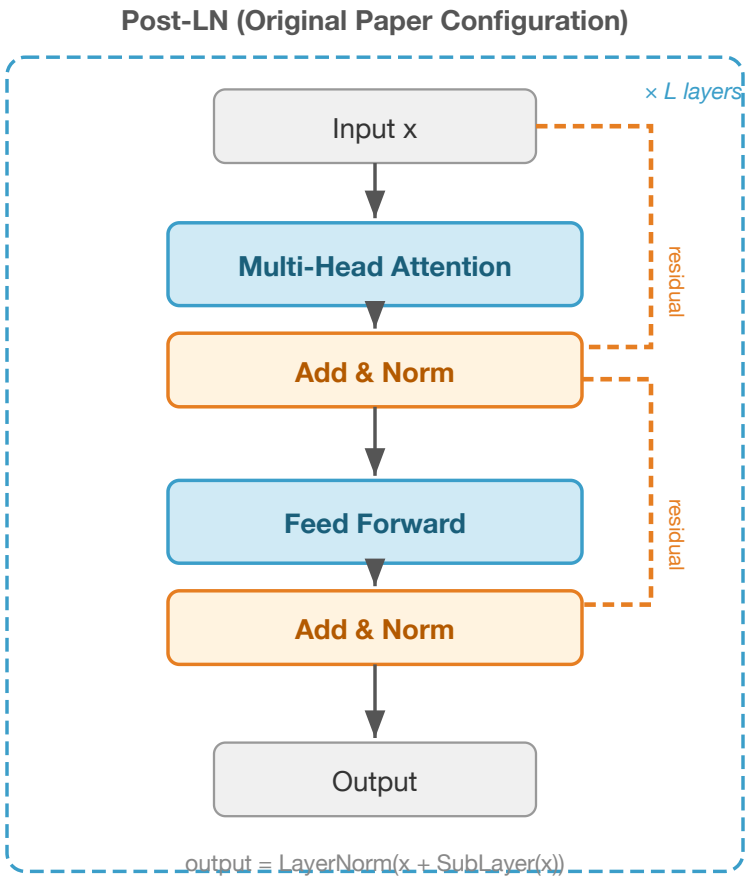
— Adds input directly (shortcut), enabling stable deep learning

Layer Normalization:

$\text{LayerNorm}(x) = \frac{x-\mu}{\sigma} \cdot \gamma + \beta$

— Normalizes per token (mean 0, var 1). γ, β learnable. Stabilizes training

BERT: **Post-LN** (Add → LN). GPT-2+: **Pre-LN** (LN → SubLayer → Add)



He et al., "Deep Residual Learning for Image Recognition", CVPR, 2016; Ba et al., "Layer Normalization", 2016

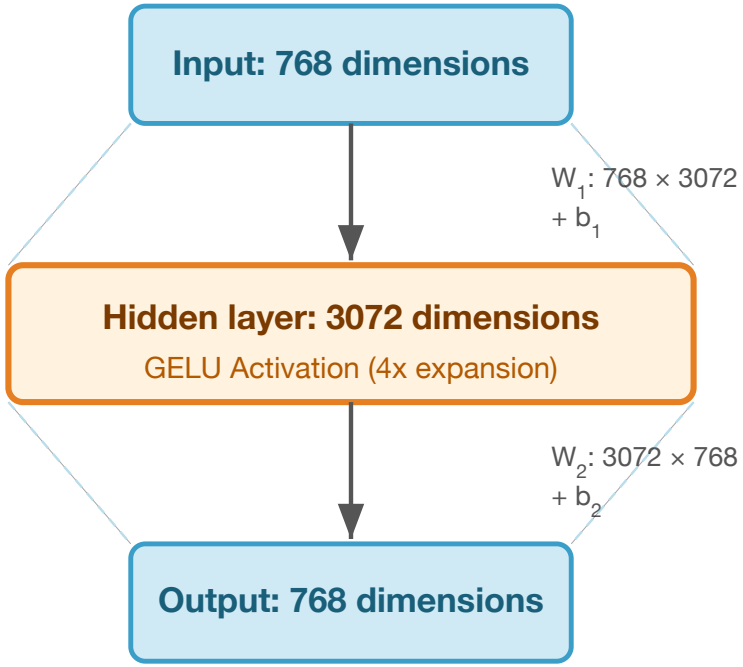
Role of the Feed-Forward Network (FFN)

A 2-layer fully connected network applied **independently to each token**

$$\mathrm{FFN}(x) = \mathrm{GELU}(xW_1 + b_1)W_2 + b_2$$

Input → intermediate ($d_{ff} = 3072$, 4× expansion) → output (768 dims)

Role: Self-Attention determines "what to gather"; FFN determines " **how to transform** " the gathered information. FFN also functions as a **knowledge memory store**



$\mathrm{FFN}(x) = \mathrm{GELU}(x\,W_1 + b_1)\,W_2 + b_2$

Applied independently to each token position

Parameters per layer in BERT-Base:
Self-Attention ≈ **2.4M** / FFN ≈ **4.7M** (about 2/3 of the total)

Geva et al., "Transformer Feed-Forward Layers Are Key-Value Memories", EMNLP, 2021

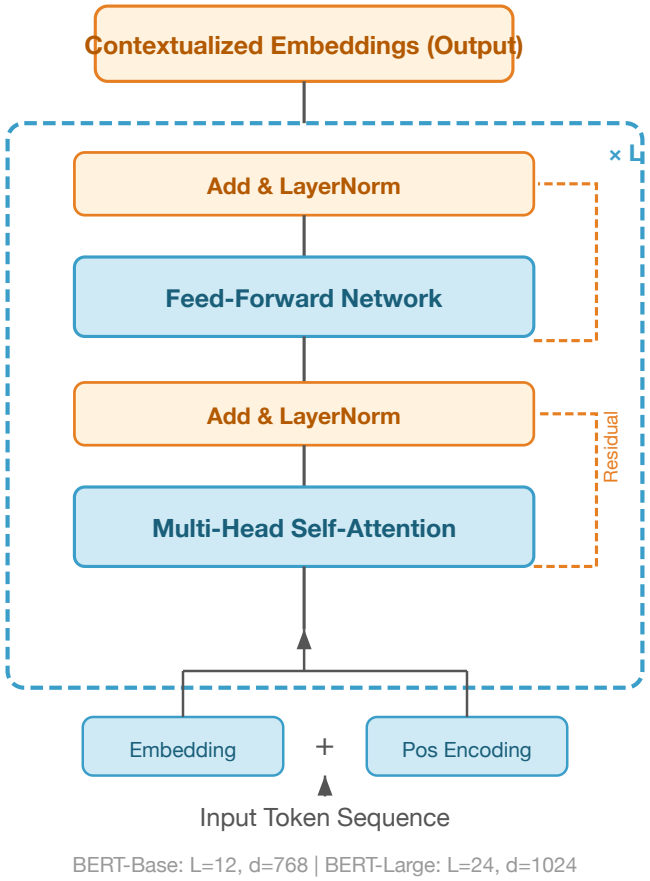
Transformer Encoder — Overall Structure

One Encoder layer = 2 sub-layers + Residual + LayerNorm:

1. **Multi-Head Self-Attention**: $z = \mathrm{LN}(x + \mathrm{MultiHead}(x))$
2. **Feed-Forward Network**: $\mathrm{out} = \mathrm{LN}(z + \mathrm{FFN}(z))$

Stacked for **L layers** (lower → syntax, upper → semantics):

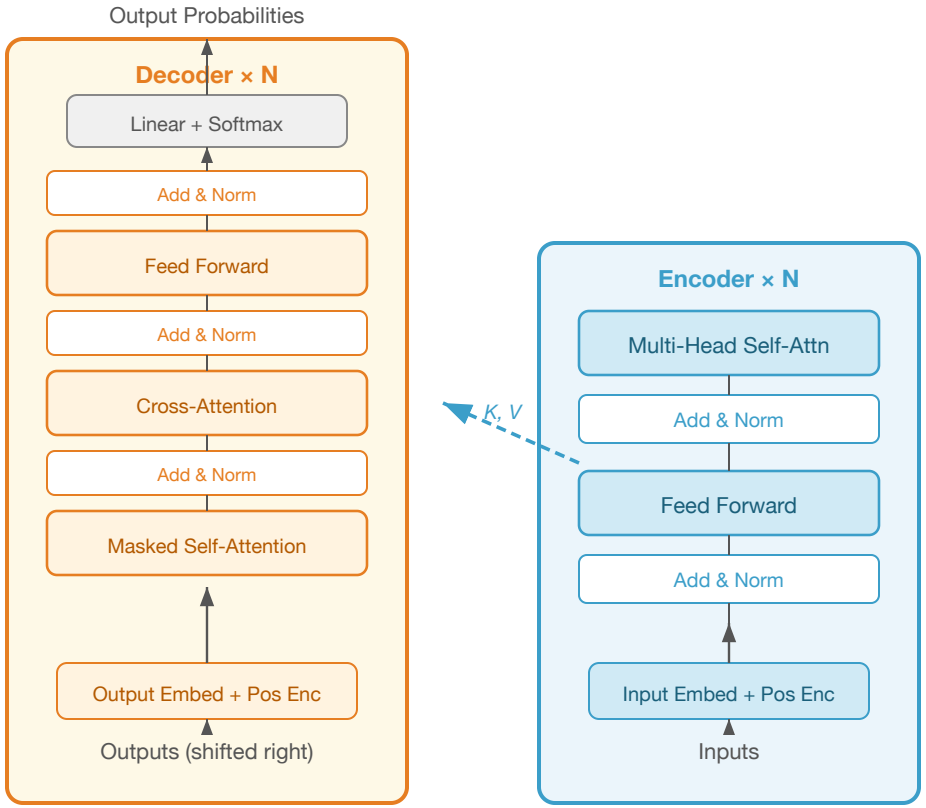
Model	L	Hidden	Heads	Params
Base	12	768	12	110M
Large	24	1024	16	340M



BERT-Base: L = 12, BERT-Large: L = 24

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

Full Picture of the Transformer — Encoder and Decoder



Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

The original Transformer has an **Encoder-Decoder** structure

Component	Feature	Derivative
Encoder (blue)	Bidirectional Self-Attn	**BERT**
Decoder (orange)	Masked + Cross-Attn	**GPT**
Both	Enc-Dec	**T5**

Each sub-layer has **Add & LayerNorm** (residual connection)

Usage in retrieval:

BERT → Cross-encoder reranking

GPT → Listwise reranking

T5 → monoT5 / duoT5

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

Comparison with pre-Transformer models:

Property	RNN/LSTM	CNN	Transformer
Capturing long-range dependencies	Difficult (gradient vanishing)	Limited by kernel size	**Direct reference (O(1) path)**
Parallel computation	Impossible (sequential)	Possible	**Fully parallel**
Bidirectional context	Possible with BiLSTM but slow	Possible	**Naturally achieved via Self-Attention**
Computational complexity	$O(n \cdot d^2)$	$O(k \cdot n \cdot d^2)$	$O(n^2 \cdot d)$

Transformer complexity $O(n^2 \cdot d)$: Expensive for long sequences. BERT max 512 tokens. Efficient variants: Longformer, BigBird

Transformer = Self-Attention + Positional Encoding + Residual Connection + LayerNorm + FFN. Each component was known, but their integration revolutionized NLP

Vaswani et al., "Attention Is All You Need", NeurIPS, 2017

BERT

Bidirectional Encoder Representations from Transformers

The Emergence of the Pre-Training Paradigm in NLP

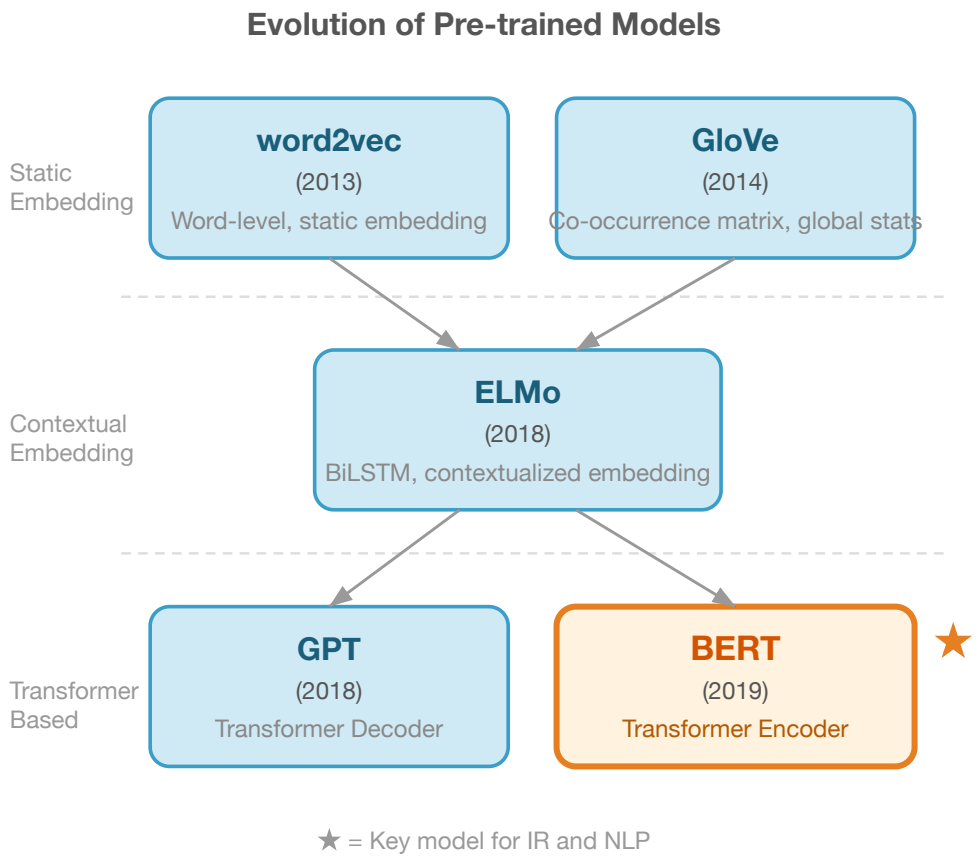
Traditional approach (~2017):

Train from scratch for each task; requires large labeled data; no knowledge sharing

Pre-training paradigm (2018~):

1. Learn general language knowledge from **large-scale unlabeled data** (pre-training)
2. Fine-tune with **small task-specific data** (fine-tuning)

Why it works: Language structure (grammar, semantics, world knowledge) is **task-independent** — once learned, it transfers to any task



Mikolov et al., 2013; Pennington et al., 2014; Peters et al., 2018; Radford et al., 2018; Devlin et al., 2019

BERT [Devlin et al., 2019] = A **contextualized embedding model** that leverages the Encoder portion of the Transformer

The innovation was combining the Transformer Encoder with **self-supervised learning**

Input: token sequence → Output: **context-dependent** representation for each token

Differences from word2vec / GloVe:

word2vec: "bank" always has the same vector

BERT: "river bank" and "bank account" → **different vectors**

BERT Model Sizes

Config	Layers	Hidden dim	Heads	Parameters
Base	12	768	12	110M
Large	24	1024	16	340M

Google released pre-trained models and code → rapid community adoption. Hugging Face Transformers is the de facto standard

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019

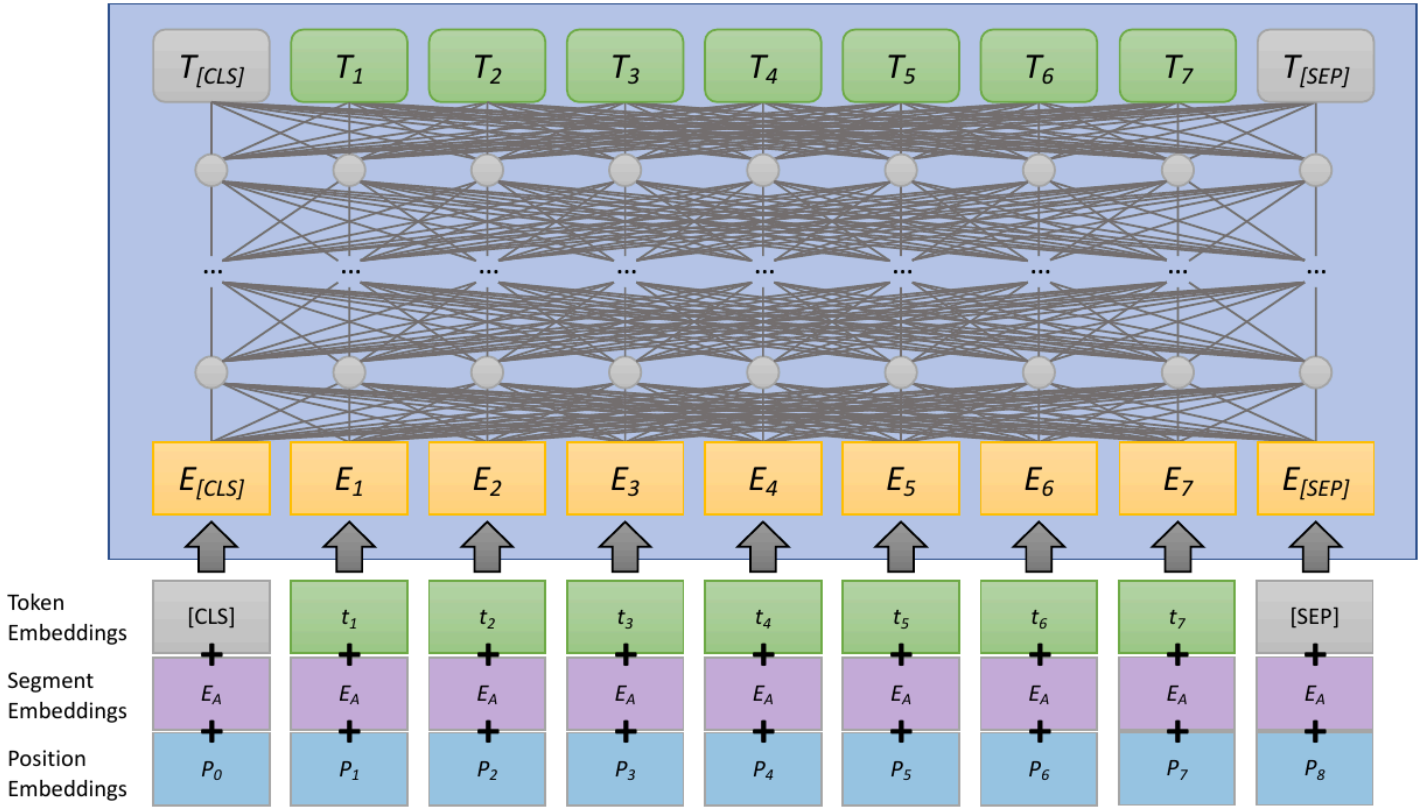
BERT uses the Transformer Encoder **as is**:

1. Convert input tokens into **embeddings**
2. **Contextualize** through L Transformer Encoder layers
3. Output **contextualized embeddings** for each token

   Input: `[CLS] t₁ t₂ ... tₙ [SEP]` →

   Output: h_i per token (768 dims)

   [CLS] token: Aggregate representation of the entire sequence, used for classification and ranking



Source: Lin et al. (2021), Figure 4

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019

BERT Input Representation

Each token's input representation is the **sum of three embeddings**:

1. **Token Embedding**: Subword embedding from the WordPiece tokenizer
2. **Segment Embedding**: Indicates whether the input belongs to Segment A or B
3. **Position Embedding**: Token position information (learnable)

Special tokens:

`[CLS]`: Aggregate representation of the entire sequence (used in classification tasks)

`[SEP]`: Separator between two inputs

Composition of Input Representation

Token	[CLS]	Kyo	to	[SEP]	au	tumn	[SEP]
Segment	A	A	A	A	B	B	B
Position	0	1	2	3	4	5	6

The three embeddings are combined by **element-wise addition** (not concatenation). The maximum sequence length is 512 tokens

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019

WordPiece Tokenization

A subword segmentation that keeps vocabulary size at approximately **30,000** while enabling representation of any text

Eliminates the unknown word problem: low-frequency words are split into subwords

The `##` prefix indicates concatenation with the preceding token

Segmentation examples:

Input word	WordPiece
scrolling	scroll + ##ing
biking	bi + ##king
information	information (no split)

Impact on IR

WordPiece has **no relation** to linguistic morphemes

"biostatistics" → "bio" + "##sta" + "##tist" + "##ics"

(does not align with morphemes)

Implications for retrieval:

BERT processes text at the **subword level**. Word-level meaning is recovered from contextualized embeddings of subwords

The 512-token limit is a limit on **subword count**, not word count

Wu et al., "Google's Neural Machine Translation System", 2016; Sennrich et al., "Neural Machine Translation of Rare Words with Subword Units", ACL, 2016

BERT Pre-Training (1): Masked Language Model

MLM (Masked Language Model):

Randomly mask **15%** of input tokens and predict the original tokens

Learns the structure of language by solving "fill-in-the-blank" problems

$$\mathcal{L}_{\text{MLM}} = -\sum_{i \in \text{masked}} \log P(t_i \mid \text{context})$$

Masking strategy (for the 15% of tokens selected for masking):

80%: Replace with `[MASK]` token

10%: Replace with a random token

10%: Leave unchanged (as is)

MLM Example

Input:

"The [MASK] leaves in Kyoto are best in November"

Prediction:

P(autumn | context) = 0.72

P(cherry | context) = 0.15

P(flower | context) = 0.08

...

What is learned:

Relationship between word meaning and context

Context-dependent usage of polysemous words

Syntactic constraints

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019

Why MLM Is Effective for Language Understanding

Correctly predicting masked words requires **deep linguistic knowledge**:

Learning syntax:

"The cat [MASK] the fish" → A verb is needed (part-of-speech understanding)

"She [MASK] the book" → "read", "bought" (argument structure understanding)

Learning semantics:

"The doctor [MASK] the patient" → "examined", "treated" (conceptual relationships)

"Tokyo is the [MASK] of Japan" → "capital" (world knowledge)

Pragmatic knowledge:

Inferring appropriate words from context requires co-occurrence patterns, causal relationships, and common-sense knowledge

Hierarchy of Knowledge Learned by MLM

Lower layers (1-4)

Part-of-speech, local syntactic patterns
"[MASK] runs" → A noun is needed

Middle layers (5-8)

Dependency parsing, semantic roles
"The doctor [MASK] the patient" → "treats"

Upper layers (9-12)

Task-specific semantic representations
Sentence-level meaning, entailment relations

Each layer of BERT progressively extracts information at higher levels of abstraction (confirmed by probing studies)

BERT Pre-Training (2): Next Sentence Prediction

NSP (Next Sentence Prediction):

- A binary classification task predicting whether two sentences are consecutive
- Determined from the output of the `[CLS]` token

Training data composition:

- 50%: Actually consecutive pairs (positive)
- 50%: Random pairs (negative)

Purpose: Acquire inter-sentence understanding (important for retrieval)

NSP Examples

IsNext (Positive)

A: "Kyoto is the ancient capital of Japan"
B: "Many temples and shrines remain there"

NotNext (Negative)

A: "Kyoto is the ancient capital of Japan"
B: "Research on quantum computers is progressing"

RoBERTa questioned NSP's necessity, but it likely helped BERT's sentence-pair understanding

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019; Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach", 2019

BERT Pre-Training Data and Computational Cost

Pre-training corpus (~3.3B words total):

- **BooksCorpus**: ~800M words (11K books)
- **English Wikipedia**: ~2.5B words

Training: Batch 256, seq len 512, 1M steps, Adam (lr 1e-4)

Computational cost:

- BERT-Base: TPU v3 × 4 pods × 4 days
- BERT-Large: TPU v3 × 16 pods × 4 days
- Estimated: **Thousands to tens of thousands of dollars**

Characteristics of Pre-Training

Key point: Pre-training only needs to be done once. By using Google's published pre-trained model, only fine-tuning is needed (1 GPU x a few hours)

Pre-Training vs Fine-Tuning Cost

	Pre-training	Fine-tuning
Data size	3.3B words	Tens to hundreds of K examples
Compute	TPU x 4-16 pods	GPU x 1
Time	Several days	Several hours
Labels	Not required	Required
Runs	Once	Per task

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019

GPT vs BERT — Two Pre-Training Paradigms

BERT (Encoder)

- **Bidirectional**: Uses both left and right context
- Pre-training: MLM + NSP
- Strengths: Classification, ranking, NER
- Primary use in document ranking: **Reranking** (Cross-encoder)

GPT (Decoder)

- **Unidirectional** (left-to-right): Uses only left context
- Pre-training: Next token prediction
- Strengths: Text generation, summarization, dialogue
- Primary use in document ranking: **Listwise reranking** (RankGPT, etc.)

$$\text{GPT}: \mathcal{L}(\mathcal{U}) = \sum_i \log P(u_i \mid u_{i-k}, \ldots, u_{i-1}; \Theta)$$

GPT: next token prediction. BERT: mask prediction — different objectives, same **"pre-train then fine-tune"** paradigm

Bidirectionality advantages: "The [MASK] in Kyoto is beautiful" → BERT uses "beautiful" as context; GPT cannot

Radford et al., "Improving Language Understanding by Generative Pre-Training", 2018; Devlin et al., 2019

BERT's Four Task Formats

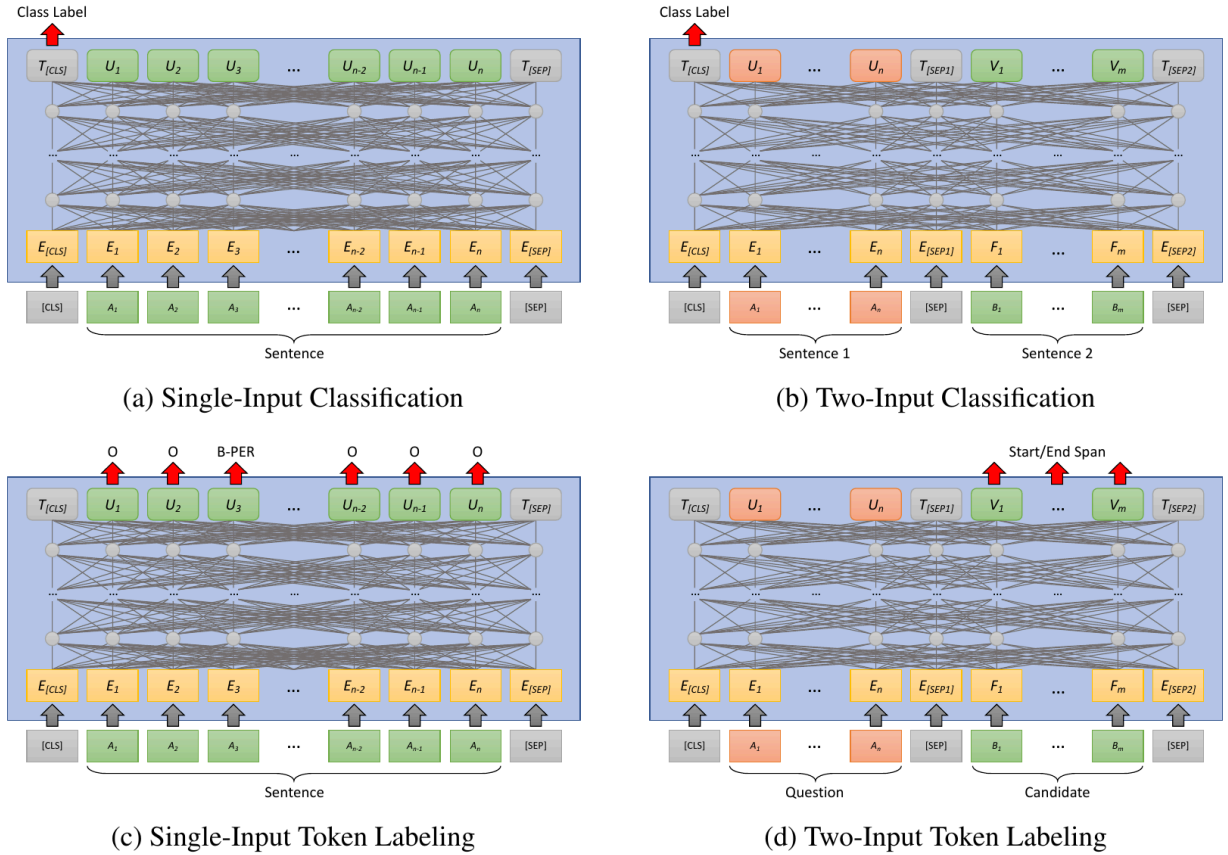
BERT handles **four task types** with the same architecture:

(a) Sentence pair classification: `[CLS]` → NLI, **document ranking**

(b) Single sentence classification: `[CLS]` → Sentiment, spam detection

(c) Question answering: Token outputs → Start/end positions (SQuAD)

(d) Token labeling: Per-token labels → NER



Source: Lin et al. (2021), Figure 5

Document ranking falls under **(a) sentence pair classification**: Input [CLS] q [SEP] d [SEP] and predict relevance from the [CLS] output. This will be covered in detail with monoBERT in the next lecture

Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding", NAACL, 2019; Lin et al. (2021), Figure 5

Impact on NLP in general:

Achieved **simultaneous** state-of-the-art on 11 NLP benchmarks (as of 2019)

GLUE benchmark: Results exceeding human performance

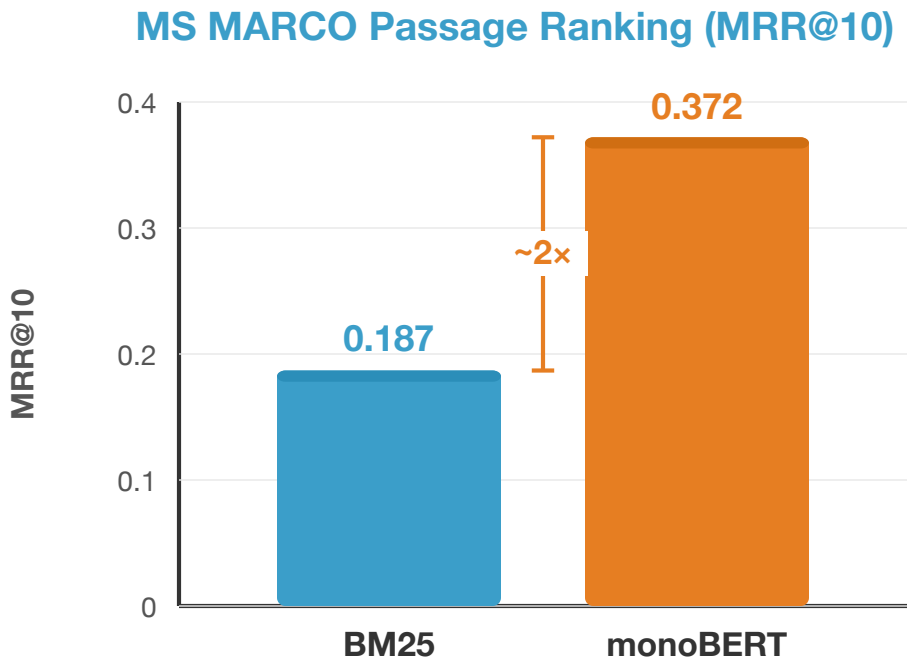
Established the paradigm of "one model for all tasks"

Impact on Information Retrieval (IR):

MS MARCO passage ranking: BM25 (MRR@10 = 0.187) → monoBERT (MRR@10 = 0.372), a **2x performance improvement**

A single model surpassed over 20 years of accumulated IR research

Standardization of the Retrieve-and-Rerank pipeline



Lin et al. (2021), Table 5

MRR@10 values from Lin et al. (2021), Table 5

Devlin et al., 2019; Nogueira and Cho, "Passage Re-ranking with BERT", 2019; Lin et al., 2021

Limitations of BERT

Architectural constraints:

- **Max 512 tokens**: Cannot process long docs → Birch, BERT-MaxP (Lecture 7)
- **Computational cost**: ~30ms/pair → impractical for full corpus
- **Cross-encoder**: Query-document concatenated → no pre-computation

Learning challenges:

- Labeled data needed for fine-tuning; domain shift degrades performance
- NSP task found unnecessary in RoBERTa

BERT Constraints and Corresponding Approaches

Constraint	Countermeasure	Lecture
512-token limit	Birch, BERT-MaxP	7
Inference speed	Knowledge distillation, multi-stage	8
Cross-encoder	Bi-encoder (DPR)	10
Lack of labeled data	doc2query, DeepCT	9
No sparse representation	SPLADE	11

Lectures 7–11 cover approaches to overcome these constraints. Understanding BERT's limitations is key to subsequent methods

Training for Ranking

Training for Ranking

Pre-Training → Fine-Tuning

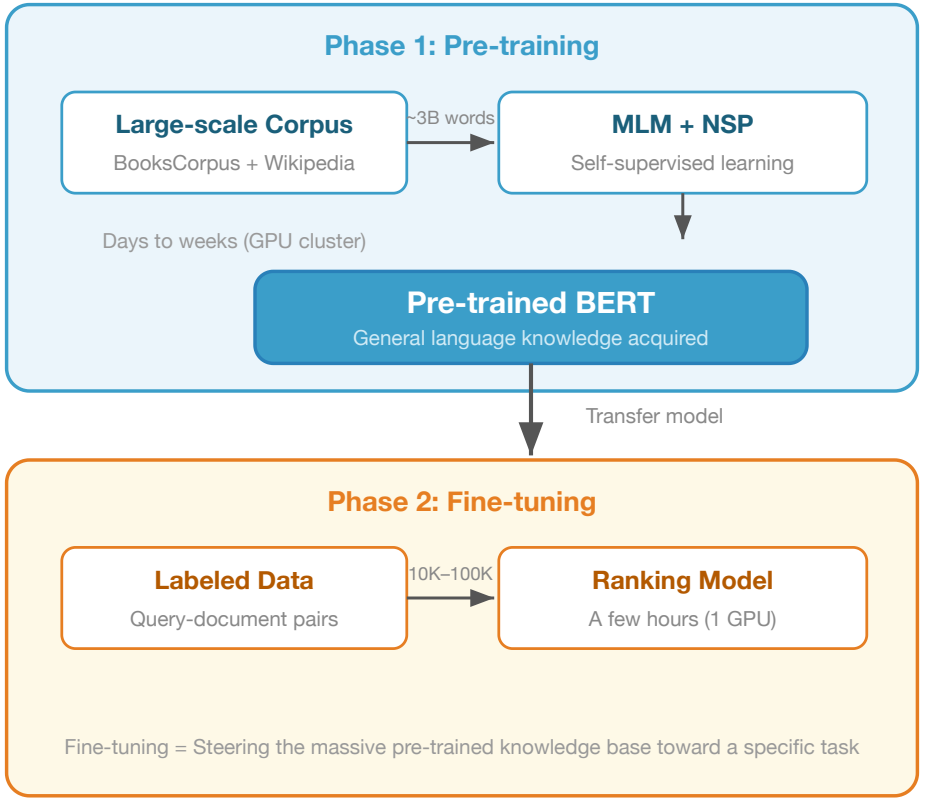
Standard recipe:

1. **Pre-training**: Learn MLM/NSP on a large corpus
2. **Fine-tuning**: Adjust on labeled data

Leverages linguistic knowledge from pre-training to build high-accuracy models with **small amounts of data**

Pre-training: Billions of words, days to weeks (GPU cluster)

FT: Tens to hundreds of K examples, several hours (1 GPU)



Devlin et al., 2019; Howard and Ruder, "Universal Language Model Fine-tuning for Text Classification", ACL, 2018

How Fine-Tuning Works — What Is Learned

Parameters being updated:

1. **All of BERT** (110M–340M): Fine-tuning the pre-trained weights
2. **Classification head** (new): Task-specific output layer

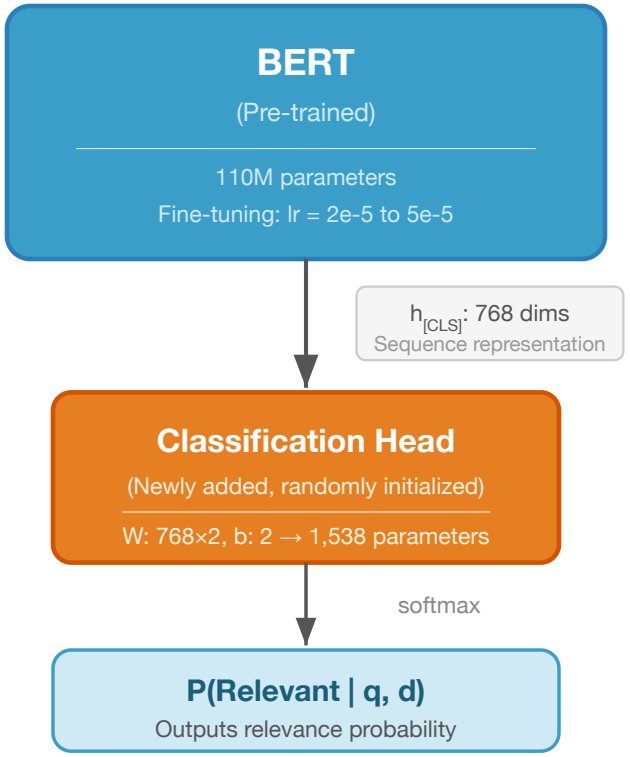
Ranking task output:

$$P(\text{Rel} \mid q, d) = \text{softmax}(h_{[\text{CLS}]} W + b)$$

$h_{[\text{CLS}]}$: [CLS] output (768 dimensions)

$W \in \mathbb{R}^{768\times 2}, b \in \mathbb{R}^2$: Additional parameters (1,538 total)

Learning rate: 2e-5 to 5e-5 (set low to protect existing weights)



99.999% of all parameters are pre-trained. Fine-tuning is the process of "directing" this massive knowledge base

Devlin et al., 2019; Lin et al., "Pretrained Transformers for Text Ranking: BERT and Beyond", 2021, § 3.2

Pointwise: Cross-Entropy Loss

The simplest approach. Binary classification of relevant/non-relevant for each (q, d) pair:

$$\mathcal{L}_{\text{CE}} = -[y \log s + (1-y)\log(1-s)]$$

$y \in \{0, 1\}$: Relevance label

$s = P(\text{Relevant} \mid q, d)$: Model output

Pairwise: Margin Loss

Training with pairs of a relevant document d^+ and a non-relevant document d^-:

$$\mathcal{L}_{\text{pair}} = \max(0, \epsilon - s(q, d^+) + s(q, d^-))$$

Comparison of Loss Functions

Loss function	Input	Characteristics
Cross-Entropy	Individual (q,d)	Simple, easy to parallelize
Pairwise Margin	(q, d^+, d^-)	Learns relative order
RankNet	(q, d^+, d^-)	Probabilistic formulation
Listwise	(q, d_1,...,d_k)	Optimizes the entire list

monoBERT uses **Pointwise CE**. Simple yet highly effective. Pairwise/Listwise often provide limited improvement relative to additional computational cost

Negative Sampling Strategies

The quality of ranking training heavily depends on **what non-relevant documents are selected**

Random negatives:

- Randomly selected from the corpus
- Most examples are too easy → Learning is inefficient

BM25 negatives:

- Non-relevant documents from the top BM25 results
- Examples that are "lexically similar but non-relevant" → More informative

Hard negatives:

- Top non-relevant documents retrieved by other models (DPR, ANCE, etc.)
- The most difficult examples to discriminate → Most effective but also risk of noise

Negative Difficulty and Learning Effect

Easy Negative (Random)

q: "Kyoto autumn leaves" / d^-: "Quantum mechanics equations"
→ Trivially non-relevant. Little learning signal

Hard Negative (BM25 top)

q: "Kyoto autumn leaves" / d^-: "Cherry blossom spots in Kyoto"
→ Lexically similar but non-relevant. Strengthens discrimination

Additional Pre-Training (Target Corpus Pretraining)

The publicly available BERT is pre-trained on a general corpus (Wikipedia + BooksCorpus)

Problem: The target retrieval corpus may differ in vocabulary distribution and genre

Target Corpus Pretraining (TCP):

- Perform additional MLM pre-training on the target retrieval corpus **before** fine-tuning
- No labeled data required (self-supervised)
- "Exposes" the model to the corpus vocabulary and style for domain adaptation

Effect of TCP (MS MARCO Passage)

Method	MRR@10
BM25	0.187
monoBERT	0.372
monoBERT + TCP	0.379

The improvement is modest, but it is a "free" improvement requiring no labels. Larger effects are expected for domain-specific settings (e.g., biomedical)

Domain-specific models such as SciBERT and BioBERT are extensions of the same idea

Nogueira et al., 2019; Lin et al., "Pretrained Transformers for Text Ranking: BERT and Beyond", 2021, § 3.2.4

Multi-Step Fine-Tuning

Strategy for when labeled data for the target task is scarce:

Pre-Fine-Tuning (pFT):

1. Start from a pre-trained BERT
2. **Pre-fine-tune** on data from a related task
3. Final fine-tuning on the target task data

Concrete examples:

MS MARCO → TREC Robust04: First learn general ranking knowledge, then adapt to a specific collection

NLI data → Ranking: Transfer knowledge of textual entailment between sentences

Overall Training Pipeline

Pre-trained BERT (general)

▼ TCP (optional)

Domain-adapted BERT

▼ pFT (optional)

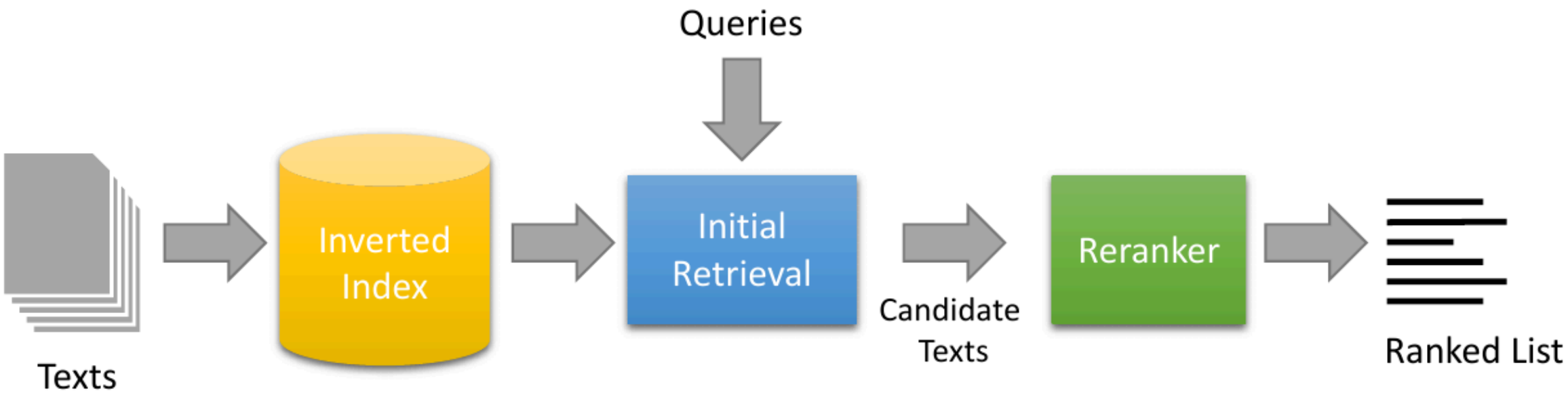
Pre-FT on related task

▼ FT

Final FT on target task

Catastrophic Forgetting: In multi-step FT, training on later datasets risks overwriting knowledge from earlier stages. BERT-based models are relatively robust, but caution is needed

Phang et al., 2018; Lin et al., "Pretrained Transformers for Text Ranking: BERT and Beyond", 2021, § 3.2.4



Source: Lin et al. (2021), Figure 6

MS MARCO passage ranking — Typical settings for monoBERT training:

1. **Pre-trained model**: `bert-base-uncased` (Hugging Face)
2. **Training data**: MS MARCO training set (~500K queries x 1 relevant passage each)
3. **Negatives**: Sample non-relevant passages from BM25 top-1,000
4. **Loss function**: Pointwise Cross-Entropy
5. **Optimization**: Adam, learning rate 3e-6, warmup 10%
6. **Batch size**: 32 / **Epochs**: 1-3 (watch for overfitting)

Nogueira and Cho, "Passage Re-ranking with BERT", 2019; Lin et al., 2021, § 3.2

1. **Neural network fundamentals**: Multi-layer perceptrons, activation functions, and embeddings represent words as dense vectors. Evolution from word2vec's static representations to contextualized embeddings
2. **Transformer**: Self-Attention captures context across the entire sequence simultaneously. The combination of Multi-Head Attention, Positional Encoding, residual connections, LayerNorm, and FFN was revolutionary
3. **BERT**: Transformer Encoder + self-supervised pre-training via MLM/NSP. Contextualized embeddings enable polysemy resolution and semantic matching
4. **Fine-tuning**: Specializes a pre-trained model for ranking tasks. Pointwise Cross-Entropy loss, BM25 negative sampling, and TCP are the fundamental techniques

Next Lecture Preview: Neural Ranking Models — Long Document Ranking

Design and ablation experiments of monoBERT

Why BERT is effective for retrieval

Handling long documents: Birch, BERT-MaxP